

## Circuit Simulation Lab:22

### Design of Counters

#### Introduction

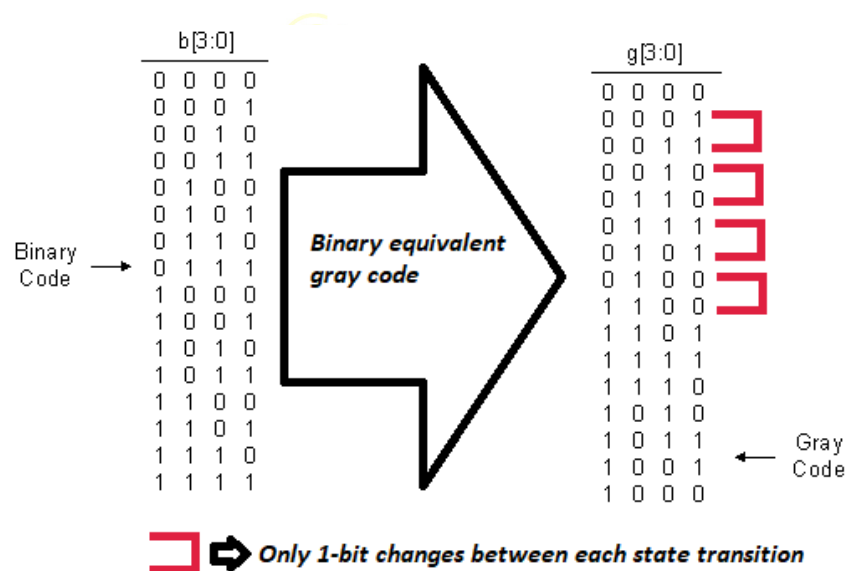
The purpose of this experiment is to introduce the design of Gray Counter.

A gray counter changes 1-bit only during one state to another state transition. The counter is same like the normal incremental counter. The only difference is in binary representation.

#### Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado



*Design a 4 bit Gray counter in Verilog HDL and write its testbench and capture the Waveforms,*

*Use Behavior modeling or Structural Modeling*